

Лабораторна робота 2

SQL: SELECT, WHERE, LIMIT, NULL, складені умови — варіант 5 (таблиця cars2)

Структура таблиці cars2

Стовпці: id (INTEGER), name (TEXT), brand (TEXT), year (INTEGER), type (TEXT), price (INTEGER), pride (TEXT, відповідає полю rating).

Ім'я	Тип
id	INTEGER
name	TEXT
brand	TEXT
year	INTEGER
type	TEXT
price	INTEGER
pride	TEXT

Завдання 1. SELECT з явним переліком стовпців

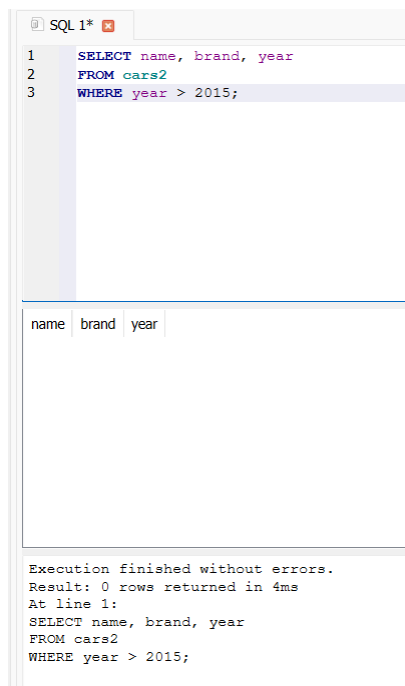
```
SELECT name, brand, year, price
FROM cars2
LIMIT 10;
```

```
Execution finished without errors.
Result: 0 rows returned in 4ms
At line 1:
SELECT name, brand, year, price
FROM cars2
LIMIT 10;
```

Завдання 2. WHERE за умовою свого варіанта

```
SELECT name, brand, year
FROM cars2
WHERE year > 2015;
```

```
Execution finished without errors.
Result: 0 rows returned in 4ms
At line 1:
SELECT name, brand, year, price
FROM cars2
LIMIT 10;
```



The screenshot shows a SQL IDE window titled "SQL 1*" with a query editor and a results pane. The query editor contains the following SQL code:

```
1 SELECT name, brand, year
2 FROM cars2
3 WHERE year > 2015;
```

The results pane shows a table with the following columns: name, brand, year. The table is currently empty, indicating that no rows were returned by the query.

Below the results pane, the execution status is displayed:

```
Execution finished without errors.
Result: 0 rows returned in 4ms
At line 1:
SELECT name, brand, year
FROM cars2
WHERE year > 2015;
```

Завдання 3. LIMIT

```
SELECT name, brand, year
FROM cars2
LIMIT 5;
```

```
SQL 1
1 SELECT name, brand, year
2 FROM cars2
3 LIMIT 5;
```

name	brand	year
------	-------	------

Execution finished without errors.
Result: 0 rows returned in 6ms
At line 1:
SELECT name, brand, year
FROM cars2
LIMIT 5;

Завдання 4. NULL, IS NULL / IS NOT NULL

```
INSERT INTO cars2 (id, name, brand, year, type, price, pride)
VALUES (1, 'Model X', 'Tesla', 2022, 'SUV', 45000, NULL);
```

```
SQL 1*
1 INSERT INTO cars2 (id, name, brand, year, type, price, pride)
2 VALUES (1, 'Model X', 'Tesla', 2022, 'SUV', 45000, NULL);
```

Execution finished without errors.
Result: query executed successfully. Took 0ms, 1 рядків постраждало
At line 1:
INSERT INTO cars2 (id, name, brand, year, type, price, pride)
VALUES (1, 'Model X', 'Tesla', 2022, 'SUV', 45000, NULL);

```
SELECT name, brand, pride
FROM cars2
WHERE pride IS NULL;
```

```
1 SELECT name, brand, pride
2 FROM cars2
3 WHERE pride IS NULL;
```

	name	brand	pride
1	Model X	Tesla	NULL

Execution finished without errors.
Result: 1 rows returned in 7ms
At line 1:
SELECT name, brand, pride
FROM cars2
WHERE pride IS NULL;

```
SELECT name, brand, pride
FROM cars2
WHERE pride IS NOT NULL;
```

```
1 SELECT name, brand, pride
2 FROM cars2
3 WHERE pride IS NOT NULL;
```

	name	brand	pride
--	------	-------	-------

Execution finished without errors.
Result: 0 rows returned in 8ms
At line 1:
SELECT name, brand, pride
FROM cars2
WHERE pride IS NOT NULL;

Чому не можна писати **= NULL**: NULL позначає не конкретне значення, а відсутність/невідомість значення. Оператор "=" порівнює два визначені значення, а порівняння з невідомим завжди дає невизначений результат (ні TRUE, ні FALSE), тому рядок ніколи не потрапить у вибірку. Для перевірки відсутності значення SQL має окремі оператори IS NULL та IS NOT NULL.

Завдання 5. Складена умова

```
SELECT name, brand, year, price
FROM cars2
WHERE year > 2015 AND price < 50000;
```

```
1 SELECT name, brand, year, price
2 FROM cars2
3 WHERE year > 2015 AND price < 50000;
```

	name	brand	year	price
1	Model X	Tesla	2022	45000

Execution finished without errors.
Result: 1 rows returned in 6ms
At line 1:
SELECT name, brand, year, price
FROM cars2
WHERE year > 2015 AND price < 50000;

```
SELECT name, brand, year, price
FROM cars2
WHERE (year > 2015 AND price < 50000) OR year > 2023;
```

```
1 SELECT name, brand, year, price
2 FROM cars2
3 WHERE (year > 2015 AND price < 50000) OR year > 2023;
```

	name	brand	year	price
1	Model X	Tesla	2022	45000

Execution finished without errors.
Result: 1 rows returned in 7ms
At line 1:
SELECT name, brand, year, price
FROM cars2
WHERE (year > 2015 AND price < 50000) OR year > 2023;